6.4.1 Condition for Parallel Lines

Lines in the same plane that do not intersect are called parallel lines.

Let $y = m_1x + c_1$ and $y = m_2x + c_2$ be the equations of two straight lines

If these two lines are parallel, then the angle between lines is zero or π (pi)

If ϕ (phi) is the angle between the lines then, $\phi = 0$

$$\Rightarrow \tan\phi = 0$$

$$\Rightarrow \frac{m_2 - m_1}{1 + m_2m_1} = 0 \Rightarrow m_2 - m_1 = 0 \Rightarrow m_2 = m_1$$

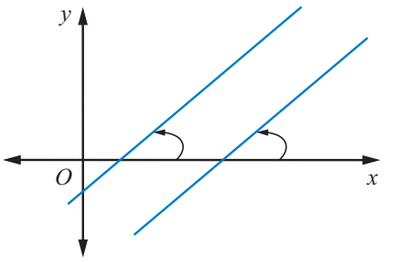


Figure 6.35

That is parallel lines have the same slope. If two non-vertical lines have the same slope, then they are parallel. All vertical lines are parallel.

If the equation of the two lines are in general form as $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$, then the condition for lines to be parallel is

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \quad \text{or} \quad a_1b_2 = a_2b_1.$$

(i) The lines parallel to $ax + by + c = 0$ are of the form $ax + by = k$.
(ii) The line parallel to $ax + by + c = 0$ and passing through a point $(x_1,\ y_1)$, then its equation is $ax + by = ax_1 + by_1$.

6.4.2 Condition for perpendicular Lines

If the lines $y = m_1x + c_1$ and $y = m_2x + c_2$ are perpendicular, then the angle between lines is $\frac{\pi}{2}$

If ϕ is the angle between the lines then

$$\tan\phi = \frac{m_2 - m_1}{1 + m_2m_1} \Rightarrow \cot\phi = \frac{1 + m_2m_1}{m_2 - m_1}$$

$$\cot\frac{\pi}{2} = \frac{1 + m_2m_1}{m_2 - m_1} \Rightarrow 1 + m_1m_2 = 0$$

$$\Rightarrow m_1m_2 = -1$$

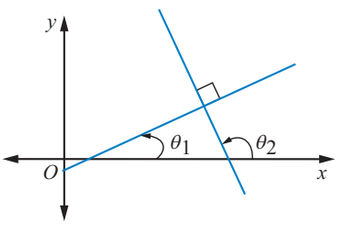


Figure 6.36

If the equation of the two lines are in general form as $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$, then the condition for lines to be perpendicular is $a_1a_2 + b_1b_2 = 0$

(i) The perpendicular line to $ax + by + c = 0$ are of the form $bx - ay = k$.
(ii) The perpendicular line to $ax + by + c = 0$ and passes through the point $(x_1,\ y_1)$, then the required equation is $bx - ay = bx_1 - ay_1$.

Form of lines	Condition for parallel	Condition for perpendicular
$y = m_1x + c_1$ and $y = m_2x + c_2$	$m_2 = m_1$	$m_1m_2 = -1$
$a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$	$a_1b_2 = a_2b_1$	$a_1a_2 + b_1b_2 = 0$